Stats

13-7-25

column => attribute, property, feature

instance/Data point/Data sample/Data example => every single row.

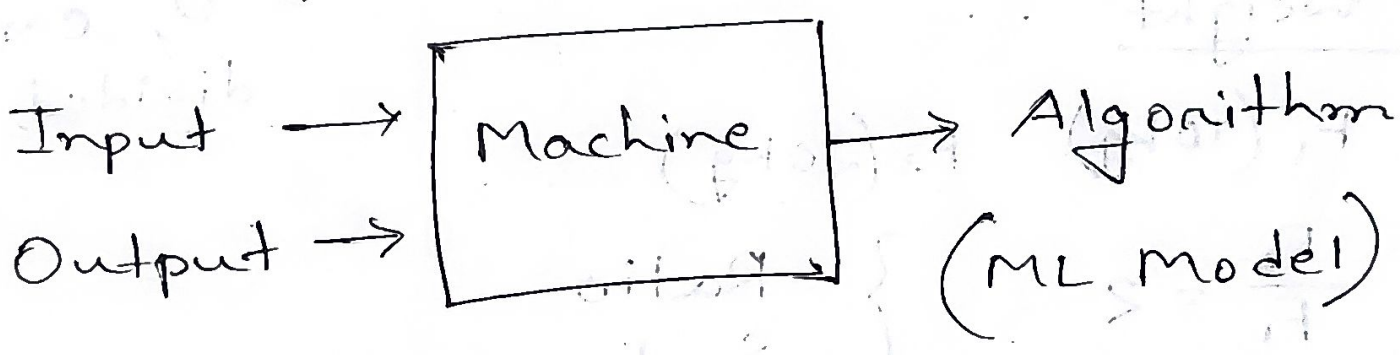
All the input column => feature

" " output " => target/label

* Supervised Learning (Labeled dataset)

* Unsupervised "

* Reinforcement " (correct, incorrect থেকে শিখবে)



Statistic

1. Collecting the dataset
2. Organization of data
3. Analyze the data
4. Interpret the data through numerical representation or visualization.

Type of data:

- Quantitative data (numerical data)
  - Discrete data
  - Continuous data
    - Interval data
    - Ratio data
- Qualitative Data (categorical data)
  - Ordinal data
  - Nominal data

Room - 103 | Room - 102

20°C | 40°C

} Interval Scale (No true 0, cannot be divided, multiplied

Weight

F_1 (10kg) F_2 (20kg)

$\frac{F_2}{F_1} = 2$ } Ratio Scale

16-7-25

Stat

output / target (Y)

Input (x)

row / column matrix can be called as vector.

$x = [Alice, 3.85, 50, 135, 165, 58, A, Advanced]$

↓

Feature vector

Supervised Learning

↓ ↓

classification Regression

Population: The entire group you want to study

Population size = N

Sample: is a subset or part of population

Sample size = n

Grade => input variable

$(x) = \{A, B, C\}$

Age = {2, 4, 4, 5, 6, 7, 7, 8, 9, 10, 10, 11, 12, 13, 13, 15}

(x_2)

$F(x_2)$

↳ Frequency of every value in x_2

min = 2

max = 15

R = [2, 15]

F(x) is a data distribution function which is used to show what are the possible values x can take and the frequency of every value.

Frequency distribution function / Data distribution function.

Grade = {A, B, C, A, A, B, B, B, C}

(x_1)

$F(x_1)$

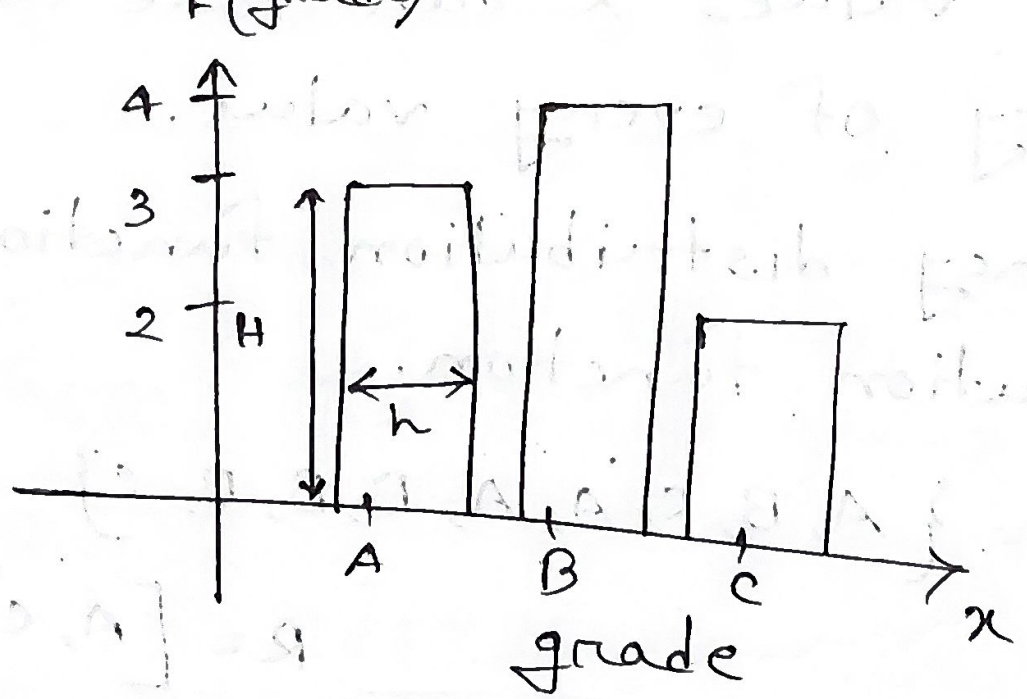
R = [A, C]

x_1	freq (x_1)
A	3
B	4
C	2

Graphical Representation of data

1) Bar graph / chart / plot (Categorical)

ii) Histogram plot (Numerical)
iii) Box plot
iv) Pie-chart
v) Steam and leaf plot
vi) Scatter plot
vii) KDE "
viii) Normal Q-Q plot



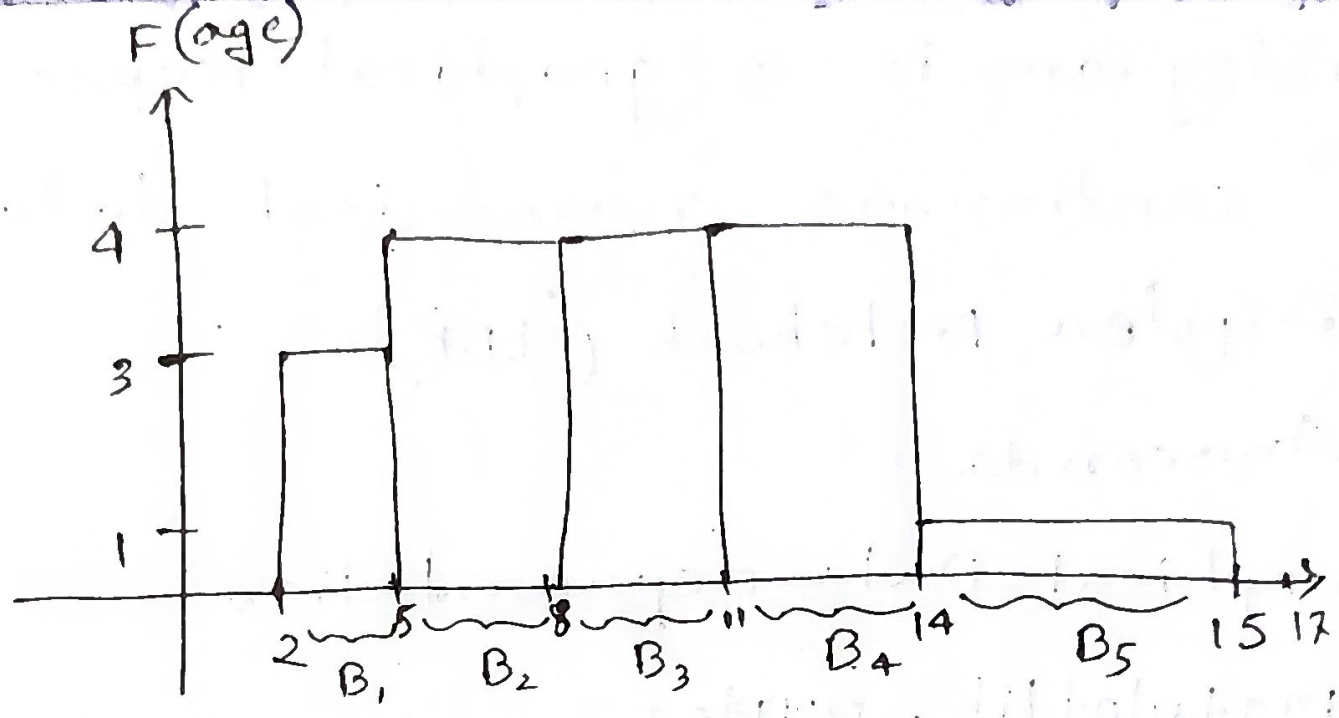
Histogram:

Step-1: $\min(x_2) = 2$

$\max(x_2) = 15$

$R = 15 - 2$

$= 13$



Step-2:

no. of intervals / bins

* Starge's Formula $[\log_2 n + 1]$
* Squaae-root choice $[\sqrt{n}]$
* Customized "

$n = 16$, $k = \lceil \log_2 n + 1 \rceil$

$= \lceil \log_2 16 + 1 \rceil = 5$

Step-3: Bin width $= \frac{R}{k} = \frac{13}{5} = \lceil 2.6 \rceil = 3$

Bin no	Bins	freq (Bins)
B_1	[2, 5)	3
B_2	[5, 8)	4
B_3	[8, 11)	4
B_4	[11, 14)	4
B_5	[14, 17)	1

* frequency represents heights of the bar.

Histogram is a graphical representat
of continuous numerical data

* Jupiter Notebook (IDE)
  Anaconda

Graphical Data representation

1) matplotlib . pyplot
2) Seaborn

plot() → line
bar()
hist()

sub-tree sum.

27-7-25

Stat

$\bar{x} = 24.375$ years

$S = 1.58$

* Given Age data set as:

$x = \{24, 25, 26, 23, 24, 22, 27, 24, 60\}$

$n = 9$

$$\bar{x} = \frac{24+25+26+23+24+22+27+24+60}{9}$$

$= 28.33$ years

$$s^2 = \frac{(24-28.33)^2 + (25-28.33)^2 + (26-28.33)^2 + (23-28.33)^2 + (24-28.33)^2 + (22-28.33)^2 + (27-28.33)^2 + (24-28.33)^2 + (60-28.33)^2}{n-1}$$

$$= 143.20 \text{ yrs}^2$$

$$s = 11.96 \text{ yrs}.$$

$[\bar{x}-s, \bar{x}+s]$

$[\bar{x}-2s, \bar{x}+2s]$

$[\bar{x}-3s, \bar{x}+3s]$

* If there is any outlayer in the DS, $\bar{x}$ does not describe the centery/location at the DS property

* Median

Step-1: Sort DS.

1	2	3	4	5	6	7	8	9 ← position/index/rank
22	23	24	24	24	25	26	27	60

$M_1 = 23.5$ $M_2 = 24$ $M_3 = 26.5$

S.S $= n = 9$

$$mid = \frac{start + end}{2} = \frac{1+n}{2} = \frac{1}{2}(n+1)$$

$$= 5$$

$$= 50\%.(n+1)$$

Median, $M_2 = x_{mid} = 24$

lower half, upper half

* Median is not influenced by the outlier

Mode → most frequent value in the DS.

* Mean is influenced by outlier.

Bimodal DS ⇒ with 2 modes.

Multimodal DS

Unimodal DS

Mean
Median
Mode
} measure of central tendency

Median of ~~ho~~ lower half:

$$mid_{lower\ half} = \frac{1+4}{2} = 2.5$$

$$M_1 = \frac{x_{mid} + x_{mid+1}}{2} = \frac{23+24}{2} = 23.5$$

Median of upper half:

$$mid = \left\lfloor \frac{6+9}{2} \right\rfloor = \lfloor 7.5 \rfloor = 7$$

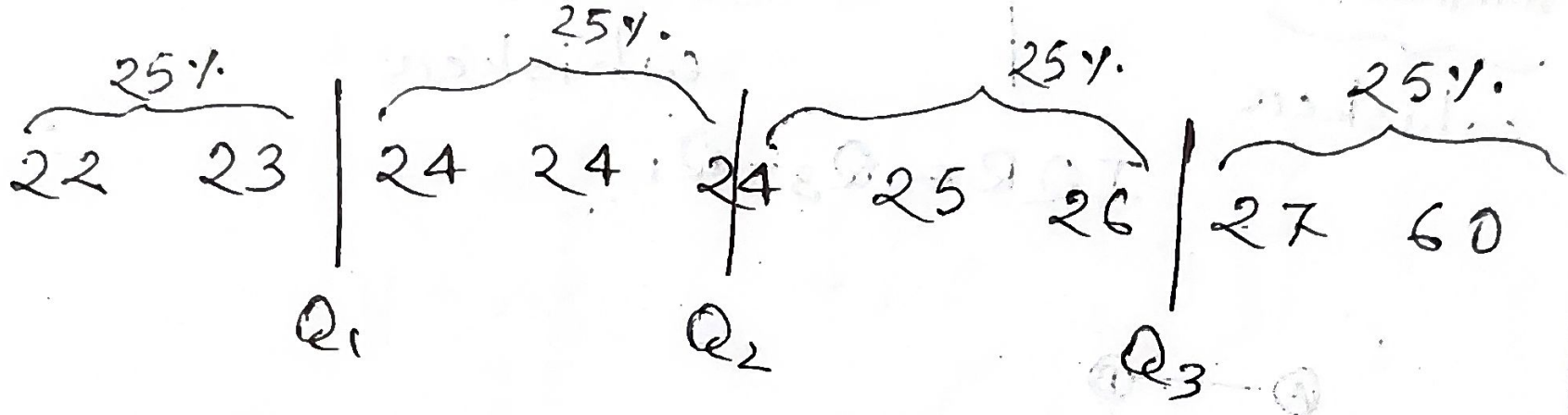
$$M_3 = \frac{x_{mid} + x_{mid+1}}{2} = \frac{26 + 27}{2} = 26.5$$

If n is even,

$$M = \frac{x_{mid} + x_{mid+1}}{2}$$

else $M = x_{mid}$

where mid = $(1+n)/2$



Inter Quartile Range,

$IQR = Q_3 - Q_1 = 26.5 - 23.5 = 3.6$ years

lower bound = $Q_1 - 1.5 \times IQR = 23.5 - 1.5 \times 3 = 19.0$

Upper " = $Q_3 + 1.5 \times IQR$

$= 26.5 + 1.5 \times 3 = 31.0$

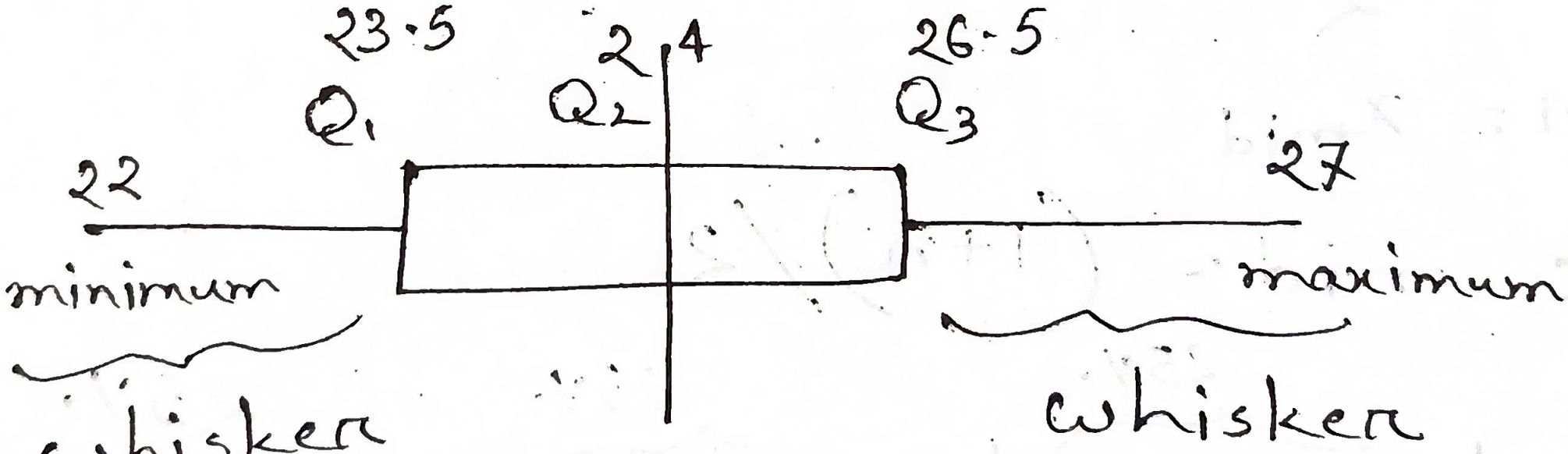
If x is any data point and if x < lower bound or x > upper bound then x is an outlier.

Outlier detection Method:

1. IQR Method / box plot.

minimum = 22

maximum = 27



$$IQR = Q_3 - Q_1$$

In any DS, P_k is the k^{th} percentile value that k% of the data are less than or equal to x.

k = percentile rank

percentile rank ≠ rank/position

Q Given a DS as,

x: 5, 7, 10, 12, 15, 14, 30, 25, 18, 21

i) Compute P_{50} $\qquad P_k = x_r$

ii) " P_{70} [Seventeeth percentile of given DS]

Step-1:

1	2	3	4	5	6	7	8	9	10	→ r
5,	7,	10,	12,	14,	15,	18,	21,	25,	30,	35

$$r = \frac{1+n}{2} = \frac{1}{2} \times (1+n) = 50\% \text{ of } (1+n)$$

$$\Rightarrow \boxed{r = k\% \text{ of } (1+n)} \Rightarrow \text{Rank computation formula}$$

$$r = \frac{k}{100} \text{ of } (1+n)$$

$$r = \frac{50}{100} \text{ of } (1+10) = \cancel{5.5}\ 6$$

$$P_{50} = x_r = x_6 = 15$$

Nearest rank formula: $r = \frac{k}{100} \times n$

ii) $k = 70$,

$$r = \frac{70}{100} \times (1+11)$$

$$= 8.4$$

~~$P_{70} = r_{lower} = r_8 = 21$~~

$$P_{70} = x_{r\text{-lower}} = x_8 = 21$$

$$P_{70} = x_{r\text{-upper}} = x_9 = 25$$

$$P_{70} = \frac{x_{r\text{-lower}} + x_{r\text{-upper}}}{2}$$

$$= \frac{x_8 + x_9}{2} = \frac{21+25}{2} = 23$$

$$P_k = x_8 + 0.4(x_9 - x_8)$$

$$= x_{[r]} + d(x_{[r]+1} - x_{[r]})$$ [linear interpolation method]

d = fractional part of r

kth- Percentile (x, k

1. Sort x in ascending order.
2. Compute $r = \frac{k}{100} \times (1+n)$
3. If r is an integer number:

$$P_k = x_r$$

else

$$d = h - \lfloor h \rfloor$$

$$P_k = x_{\lfloor h \rfloor} + d\left(x_{\lfloor h \rfloor + 1} - x_{\lfloor h \rfloor}\right)$$

or,

$$P_k = x_{\lfloor h \rfloor}$$

or, $P_k = x_{\lceil h \rceil}$

or, $$P_k = \frac{x_{\lfloor h \rfloor} + x_{\lceil h \rceil}}{2}$$

iv. Return P_k

(iii) What percentile of the DS is 30?

$\frac{10}{11} \times 100\% = 90.9\%$

Percentile rank, $k = \frac{\text{number of values} \leq x_n}{\text{Total no. of values / Sample size}} \times 100\%$

Q

Student	Score in ~~stat~~ algo	Score in algorithm
A	78	
B	79	
C	80	
D	81	
E	82	

Range, $R_{DS-1} = 82 - 78 = 4$

DS-2

Student	Score in Stat
A	60
B	70
C	80
D	90
E	100

$R_{DS-2} = 100 - 60 = 40$

Compare DS-1 and DS-2

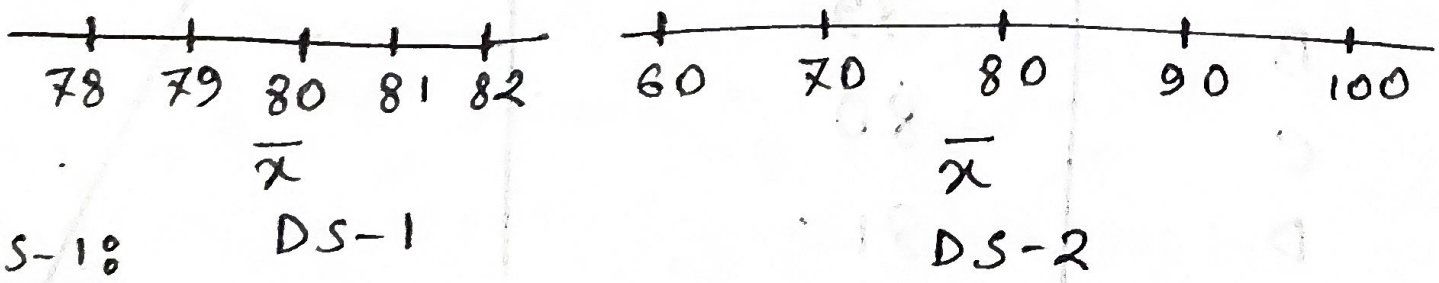
	Mean	Varience	standard deviation, s
DS-1	80.0	2.50	1.58
DS-2	80.0	250.00	15.81

↑ Statistical Summary

Bi-variate analysis

Uni- " "

Multi- " "



DS-1:

The values of DS-1 are more tightly concentrated around mean $\bar{x}$ DS-1.

The DS have smaller Variance (S.D)

For DS-2: The values are more spread out/ dispersed from the mean, DS-2.

∴ The DS has greater Varience S.D

Measure of Dispersed / Variability

1. Variance (s^2)
2. S.D (s)
3. Range, R = max - min
4. IQR

* Middle in 50% of the data of DS-1 are more or tightly concentrate around median value.

1st moment business decision

- → mean $\bar{x}$ or $E[x]$
- → median
- → mode

1st moment ⇒ μ_1

Second moment ⇒ $\mu_2 = E[(x-\mu)^2]$

⇓

Variance

Stat

31-7-25

Q Given two D.S as x and y. Describe there relationship.

$x = \{2, 4, 6, 8, 10\}$

$y = \{65, 70, 75, 80, 85\}$

Covariance, $cov(x, y)$

$y = 2x$

↑ x ↑ y } +ve
↓ x ↓ y } $cov(x, y)$

$y = \frac{1}{2}x$ $\frac{1}{x}$

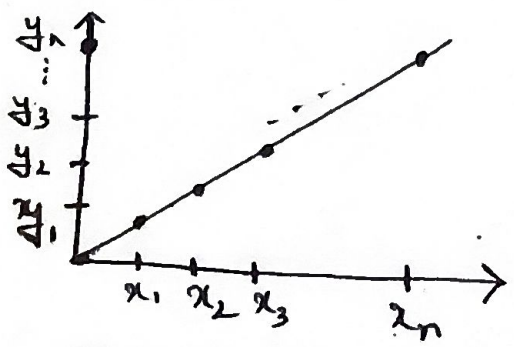
↑ x ↓ y } -ve $cov(x, y)$
↓ x ↑ y }

$cov(x, y) = 0$, there is no relation between x and y.

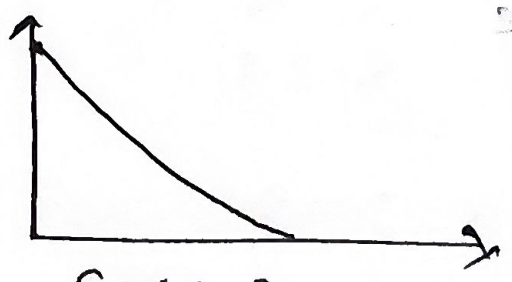
Determine Covariance of x and y.

$x = \{x_1, x_2, x_3, \ldots, x_n\}$

$y = \{y_1, y_2, y_3, \ldots, y_n\}$



Case 1



Case 2

Sample $Cov(x,y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$

Population $Cov(x,y) = \frac{1}{N}\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})$

x	y	$x-\bar{x}$	$y-\bar{y}$
2	65		
4	70		
6	75		
8	80		
10	85		

Step-1:

$$\bar{x} = \frac{2+4+6+8+10}{5} = 6$$

$$\bar{y} = \frac{65+70+75+80+85}{5} = 75$$

Step-2

x	y	$x-\bar{x}$	$y-\bar{y}$	$(x-\bar{x})(y-\bar{y})$
2	65	−4	−10	40
4	70	−2	−5	10
6	75	0	0	0
8	80	2	5	10
10	85	4	10	40
				100

$$Cov(x,y) = \frac{1}{n-1}\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})$$

$$= \frac{1}{4} \times 100$$

$$= 25 \quad \text{years} \times \text{marks}$$

There is positive relationship between x and y.

Normalized Covariance → Correlation

$x = \{x_1, x_2, x_3 \dots x_n\}$

$y = \{y_1, y_2, y_3, \dots, y_n\}$

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad \bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$$

$$S_x^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \qquad S_y^2$$

$$S_x = \sqrt{S_x^2} \qquad S_y$$

Correlation (x, y), $r_0 = \dfrac{\text{Cov}(x, y)}{S_x \cdot S_y}$

$$r_0 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}} \qquad r_0 \in [-1, 1]$$

3-8-25

$$r = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

$r = [-1, 1]$

Variation DS:

$\vec{X} = \{x_1 - \bar{x}, x_2 - \bar{x}, x_3 - \bar{x}, \ldots x_n - \bar{x}\}$

$\vec{Y} = \{y_1 - \bar{y}, y_2 - \bar{y}, y_3 - \bar{y}, \ldots y_n - \bar{y}\}$

$\vec{X} \cdot \vec{Y} = (x_1 - \bar{x})(y_1 - \bar{y}) + (x_2 - \bar{x})(y_2 - \bar{y}) + \ldots (x_n - \bar{x})(y_n - \bar{y})$

$= \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$ ——— (1)

$|\vec{X}| = \sqrt{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2}$

$= \sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}$

$$|\vec{Y}| = \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}$$

From (1) =>

$$|\vec{x}|\,|\vec{Y}| \cos\theta = \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})$$

$$\Rightarrow \cos\theta = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n} (x_i - \bar{x})^2}\ \sqrt{\sum_{i=1}^{n} (y_i - \bar{y})^2}}$$

$\therefore r = \cos\theta$

$r \in [-1, 1]$

r	Types of correlation
1	Perfect +ve
0.9 - 0.7	Strong +ve
0.6 - 0.4	Moderate +ve
0.3 - 0.1	Weak +ve
0	No correlation
-0.1 to (0.3)	Weak -ve
-0.4 to -0.6	Moderate -ve
-0.7 to -0.9	Strong -ve
-1	Perfect -ve

Ways to compute mean:

1. Arithmetic mean

Feature matrix

ML models:

* Linear Regression model
* Logistic " "
* Naive baye's classifies
* Decision tree
* SVM (Support Vector Machine)

Features	r(x, y)
House ~~Paice~~ Size	+0.95
No. of bedrooms	+0.50
Distance to city center	-0.23
Age	-0.13

Heat map:

	H.S	N.O.B	D.T.C.C	H.P
H.S	1.0	+0.97		+0.95
N.O.B		1.0	+0.23	+0.92
D.T.C.C			1	
H.P				1

* Correlation helps in feature selection

1. EDA
2. Feature Selection
3. Better model design
4. Interpretability

kendal

Pearson correlation coefficient

6-8-25

Stat

	f_1	f_2	f_3	.. f_n	$\vec{y}$
x_1					y_1
x_2					y_2
⋮					⋮
x_n					y_n

* Regression analysis :

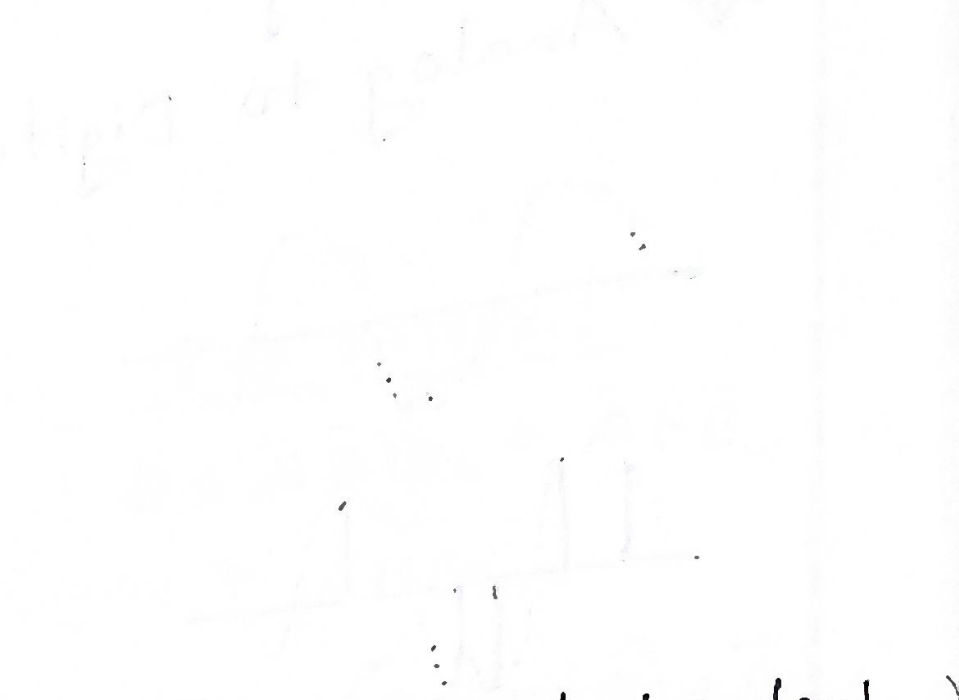
1. Simple Linear Regression (1 input col, 1 output col)
2. Multiple " "
3. Polynomial " "
4. Logistic " "
5. Ridge-Lasso "

cgpa (x)	Salary (y)
1	2
2	4
3	6
3.5	7
3.6	7.2

$\hat{y} = mx + b$

$\hat{y} = 2x + 0 = 2 \times cgpa + 0$

$\hat{y} = \omega x + b, \; b = 0$



$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4-2}{2-1} = 2$$

$b = 0$

$\hat{y}_1 = wx_1 = 2 \times 1 = 2 \quad [y_1 = \hat{y}_1]$ fit

Objective: Find out the best regression line, for which the error produced by the model is minimum.

error $= e_6 = y_6 - \hat{y}_6$

Loss, L_6 ; Residual / error

Total Loss, $L = L_1 + L_2 + L_3 + \cdots L_n$

$$= \sum_{i=1}^{n} L_i = \sum_{i=1}^{n} |(y_i - \hat{y}_i)|$$

$L_i = (y_i - \hat{y}_i)$

Cost, C / J

Mean absolute error $= \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$

Sum of squared error (SSE), $C = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

Mean squared error (MSE) $= \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$

Root " " " $= \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$

Sum of square of residuals

or

Residual Sum of Square (RSS)

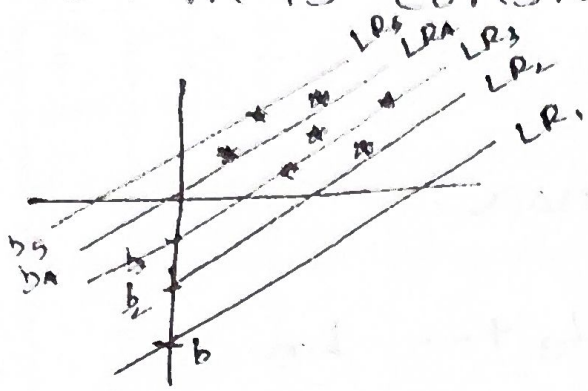
Best LR model when C is minimum

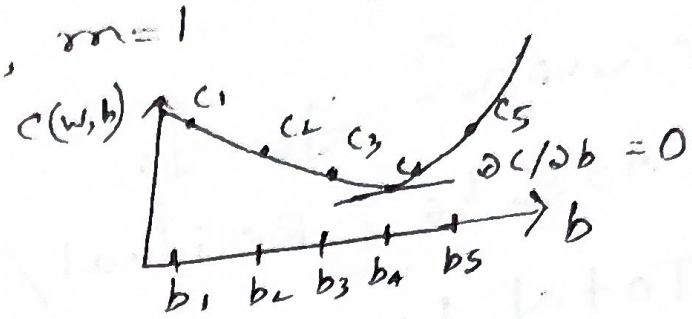
$$C = \sum_{i=1}^{n} (y_i - wx_i - b)^2 = C(w, b) \Rightarrow \text{cost function}$$

To find such a value for w and b so that $c(w,b)$ is minimum.

optimal w, b.

When m is constant, $m=1$





$$\frac{\partial c}{\partial b}$$

$$c(w,b) = \sum_{i=1}^{n} (y_i - wx_i - b)^2$$

$$\frac{\partial c}{\partial b} = \sum_{i=1}^{n} \frac{\partial}{\partial b} \left[(y_i - wx_i - b)^2 \right]$$

$$= \sum_{i=1}^{n} 2(y_i - wx_i - b) \frac{\partial}{\partial b} (y_i - wx_i - b)$$

$$= \sum_{i=1}^{n} 2(y_i - wx_i - b)(-1)$$

$$\frac{\partial c}{\partial b} = 0$$

$$\Rightarrow -2 \sum_{i=1}^{n} (y_i - wx_i - b) = 0$$

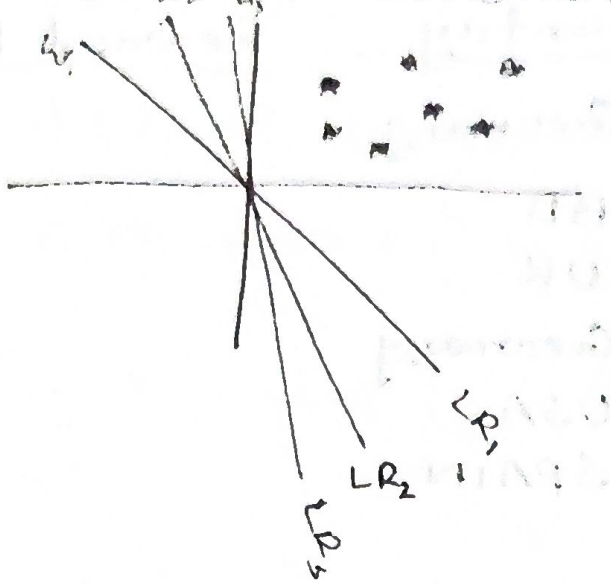
$$\Rightarrow \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} wx_i - \sum_{i=1}^{n} b = 0$$

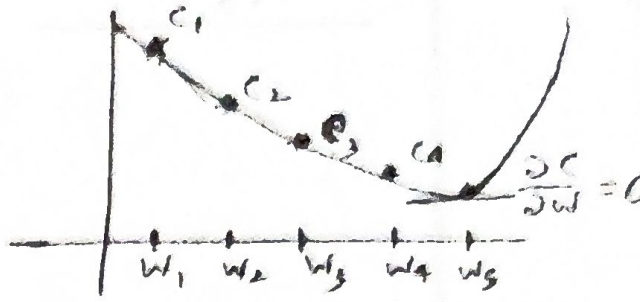
$$\Rightarrow \frac{\sum_{i=1}^{n} y_i}{n} - \frac{w \sum_{i=1}^{n} x_i}{n} - \frac{\sum_{i=1}^{n} b}{n} = 0$$

$$\Rightarrow \bar{y} - w\bar{x} - \frac{nb}{n} = 0$$

$$\Rightarrow b = \bar{y} - w\bar{x}$$

When w_1, b_1 is constant, $b = 0$





$\frac{\partial C}{\partial w} = 0$ [for optimum w]

$$\Rightarrow \frac{\partial}{\partial w}\left[\sum_{i=1}^{n}(y_i - wx_i - b)^2\right] = 0$$

$$\Rightarrow \sum_{i=1}^{n}\frac{\partial}{\partial w}(y_i - wx_i - \bar{y} + w\bar{x})^2 = 0 \qquad [b = \bar{y} - w\bar{x}]$$

$$\Rightarrow \sum_{i=1}^{n} 2(y_i - wx_i - \bar{y} + w\bar{x})(-x_i + \bar{x}) = 0$$

$$\Rightarrow -2\sum_{i=1}^{n}(y_i - wx_i - \bar{y} + w\bar{x})(x_i - \bar{x}) = 0$$

$$\Rightarrow \sum_{i=1}^{n}\{(y_i - \bar{y}) - w(x_i - \bar{x})\}(x_i - \bar{x}) = 0$$

$$\Rightarrow \sum_{i=1}^{n}\{(x_i - \bar{x})(y_i - \bar{y}) - w(x_i - \bar{x})^2\} = 0$$

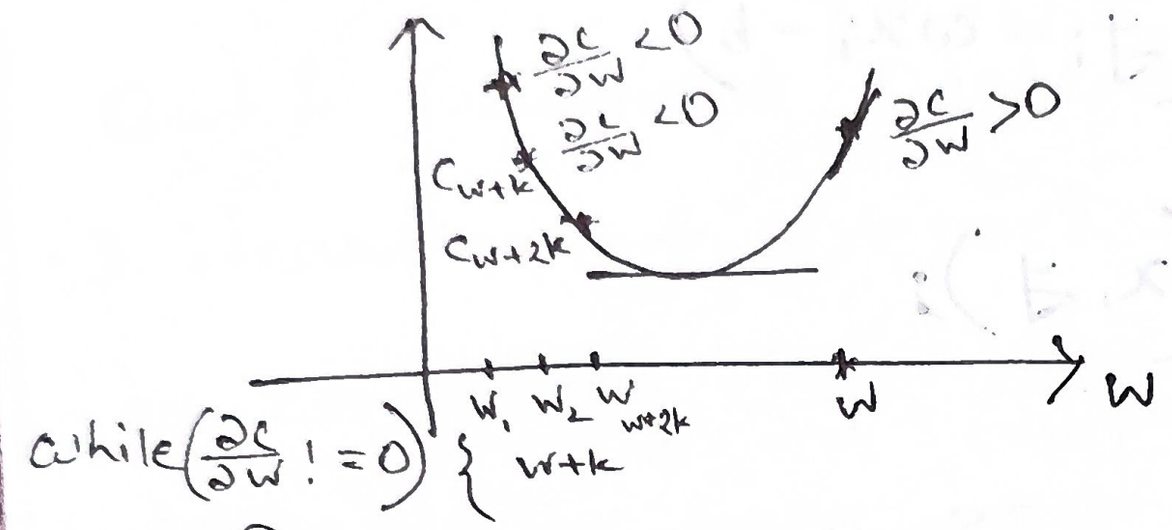
$$\Rightarrow \sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) - w\sum_{i=1}^{n}(x_i - \bar{x})^2 = 0$$

$$\Rightarrow w = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$w = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}$$

Stat

13-8-25



while $\left(\frac{\partial c}{\partial w} != 0\right)$ { $w+k$

if for some, w $\frac{\partial c}{\partial w} < 0$

$w = w + \square \eta * \frac{\partial c}{\partial w}$

if $\frac{\partial c}{\partial w} > 0$

$w = w - \square \eta * \frac{\partial c}{\partial w}$

}

$\Delta w = w_2 - w_1$

$\partial w \longrightarrow \partial c$

$1 \longrightarrow \frac{\partial c}{\partial w}$

$\eta \longrightarrow \boxed{\eta * \frac{\partial c}{\partial w}}$

gradient descent

$$C = \frac{1}{m}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2$$

$$\frac{\partial C}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}(y_i - wx_i - b)^2$$

$$= \frac{1}{m}\sum_{i=1}^{m} 2(y_i - wx_i - b)(0 - x_i - 0)$$

$$= -\frac{2}{m}\sum_{i=1}^{m}(y_i - wx_i - b)x_i$$

if for some b, $\frac{\partial C}{\partial b} < 0$,

$$b = b - \eta * \frac{\partial C}{\partial b}$$

$\frac{\partial C}{\partial b} > 0$,

$$b = b - \eta * \frac{\partial C}{\partial b}$$

$$C = \frac{1}{m}\sum_{i=1}^{m}(y_i - wx_i - b)^2$$

$$\frac{\partial C}{\partial b} = \frac{1}{m}\sum_{i=1}^{m} 2(y_i - wx_i - b)(-1)$$

$$= -\frac{2}{m}\sum_{i=1}^{m}(y_i - wx_i - b)$$

gradient_decent (x, y);

(i).

(ii). Compute $\frac{\partial C}{\partial w}$

(iii) " $\frac{\partial C}{\partial b}$

iii. Assume some random value for w and b,

iv. set $\eta = 0.1$ [learning rate]

v. for $i = 1$ to 10^7

$$w = w - \eta * \frac{\partial c}{\partial w}$$

$$b = b - \eta * \frac{\partial c}{\partial b}$$

vi. Return w, b [optimal value of w and b]

Hyperparameter (η)

gradient_decent (x, y):

(i) Assume some random value for w and b.

(ii) Set $\eta = 0.1$

(iii) for i=1 to 10^7

a. Compute $\frac{\partial c}{\partial w}$

b. " $\frac{\partial c}{\partial b}$

c. $w = w - \eta * \frac{\partial c}{\partial w}$

d. $b = b - \eta * \frac{\partial c}{\partial w}$

(iv). Return w, b [optimal value of w and b]

Batch Gradient Descent (BGD)

1 iteration → 1 epoch

for computing $\frac{\partial c}{\partial w}$, $\frac{\partial c}{\partial b}$:

sum_w = 0, sum_b = 0

for i = 1 to m

sum_b += sum_b + (y[i] - wx[i] - b)

sum_w = sum_w + (y[i] - w[xi] - b) x[i]

$\frac{\partial c}{\partial b}$ = sum_b * $(-\frac{2}{m})$

$\frac{\partial c}{\partial w}$ = sum_w * $(-\frac{2}{m})$

for individual data point,

$$\frac{\partial c}{\partial w} = -\frac{2}{m}(y_i - wx_i - b)x_i$$

$$\frac{\partial c}{\partial b} = -\frac{2}{m}(y_i - wx_i - b)$$

stochastic GD
↓
accuracy कम,
does not suffe
from time
complexity

SGD (x, y) :

1. w = rand() b = rand()
   for (i = 1 to 100) {
ii for i = 1 to m
   {
   $\partial c_\partial w = (-2/m)(y_i - wx_i - b)x_i$
   $\partial c_\partial b = (-2/m)(y_i - wx_i - b)$
   $w = w - \eta * (\partial c_\partial w)$
   $b = b - \eta * (\partial c_\partial b)$
   }
   }

mini-Batch GD => optimal

x	y	$\hat{y}$
x_1	y_1	$\hat{y}_1$
x_2	y_2	$\hat{y}_2$
...	...	...
x_i	y_i	$\hat{y}_i$
...	...	...
x_m	y_m	$\hat{y}_m$

OLS (Ordinary least square)

$$J_i / L_i / e_i = \frac{1}{m}\sum\underbrace{(y_i - \hat{y}_i)^2}_{\text{residual}}$$

RSS ⇒ Residual Sum of Squares
↳ unexplained variation

R^2 squared method

Total variation / TSS (Total Sum of Square)

$$= \frac{1}{m}\sum(y_i - \bar{y})^2$$

Explained variation = TSS - RSS

$$ESS = \frac{1}{m}\sum(\hat{y}_i - \bar{y}_i)^2$$

Coefficient of determination R^2 = $\dfrac{\text{explained variation}}{\text{Total variation}}$

$$R^2 = \frac{ESS}{TSS} = \frac{TSS - RSS}{TSS}$$

$$= 1 - \frac{RSS}{TSS}$$

$R^2 = 1$ [RSS = 0] When $\hat{y}_i = \bar{y}$ then, RSS = TSS

$R^2 < 0$ [RSS > TSS] $R^2 = 0$

R^2 ⇒ loss function / error function

x_1	x_2	x_3	x_4	x_5	x_6	x_7
4	4	5	5	5	3	8

$n = 7$

Simple / Arithmetic mean $\bar{x}$

	x	$f(x)$
x_1	3	1 ω_1
x_2	4	2 f_i/ω_i ω_2
x_3	5	3 ω_3
x_4	8	1 ω_4

$$\bar{x} = \frac{3\times1 + 4\times2 + 5\times3 + 8\times1}{1+2+3+1}$$

$$\bar{x} = \frac{x_1\times\omega_1 + x_2\times\omega_2 + x_3\times\omega_3 + x_4\times\omega_4}{\omega_1+\omega_2+\omega_3+\omega_4}$$

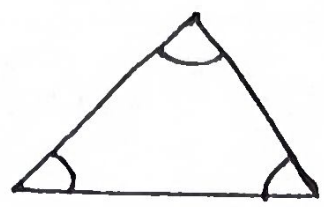
	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8
	2.4	4	3.2	4	3.9	3.95	3.76	3.3
Credit	20	17	18	22	21	17	16	20

Stat

24-8-25

* Probability => is the chance of happening something.

* Experiment



X = The sum of three angles of a triangle
= 180°

Tossing a coin (H, T)

* If the outcome of an experiment does not know in advance, it is called random experime

outcome = a single result of a R.E

* Rolling a dice

{1, 2, 3, 4, 5, 6} => Sample Space (S)

Sample Space: The ~~result~~ set of all outcomes in a R.E

X → represents any single outcome of R.A.D

Random Variable => represents numerical output.

$X = 1$ $\quad 1 = H$

$0 \quad 0 = T$

* Event => subset of sample space

$E_1 = \{H\}$

$E_2 = \{T\}$

$A = \{2, 4, 6\}$

$B = \{1, 3, 5\}$

$C = \{5, 6\}$

$D = \{6\}$

$$P(A) = \frac{3}{6} = 0.5$$

$$P(A) = \frac{\text{No of favourable outcome to A}}{\text{Total no of outcome/sample size}}$$

$$P(B) = \frac{3}{6} = 0.5$$

$$P(C) = \frac{2}{6} = 0.33$$

$$P(D) = \frac{1}{6}$$

$P(x=1) = \frac{1}{6}$ $\quad P(x=5) = \frac{1}{6}$

$P(x=2) = \frac{1}{6}$ $\quad P(x=6) = \frac{1}{6}$

$P(x=3) = \frac{1}{6}$

$P(x=4) = \frac{1}{6}$

$P(x=1) \ldots P(x=6)$ are equiprobable or they are equally likely.

$0 \leq P(A) \leq 1$

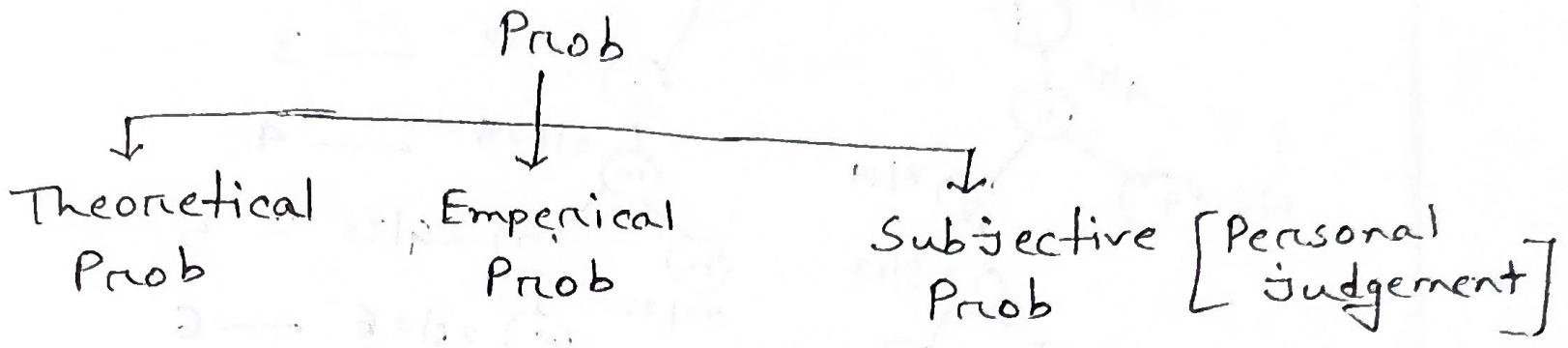
$0\% \leq P(A) \leq 100\%$

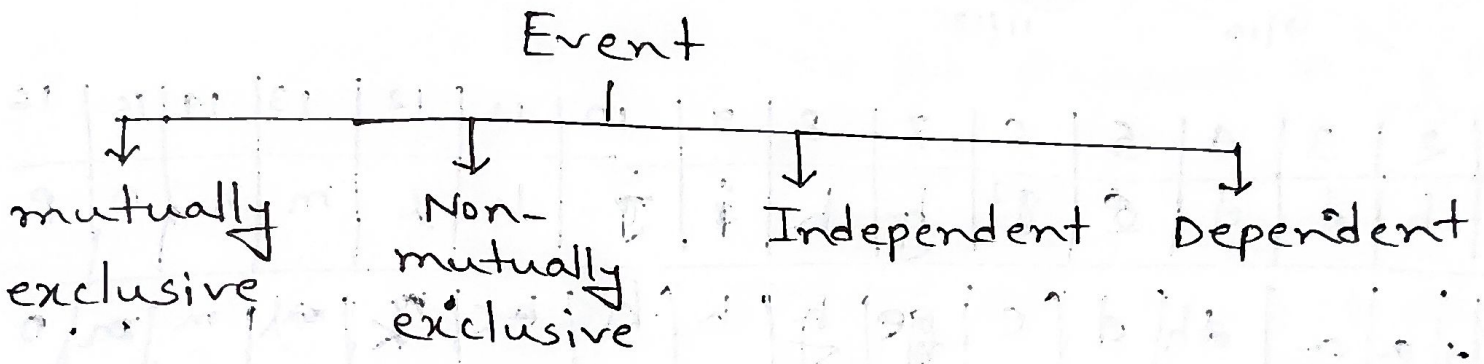
likelihood / Prob

$P(H) = 1/2$

$P(X = H) = \frac{1}{2}$

$P(T) = \frac{1}{2}$





A ↓ Even num

B ↓ Odd num

$P(A \cap B) = 0$

$P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

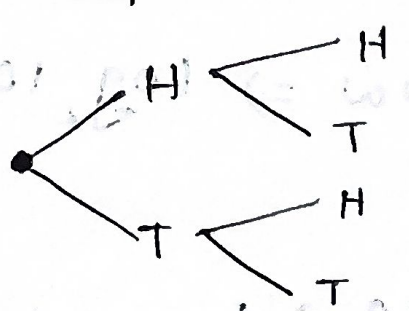
Mutually Exclusive $\Rightarrow P(A \cup B) = P(A) + P(B)$

* Tossing a coin twice

A — Toss 1: H, T

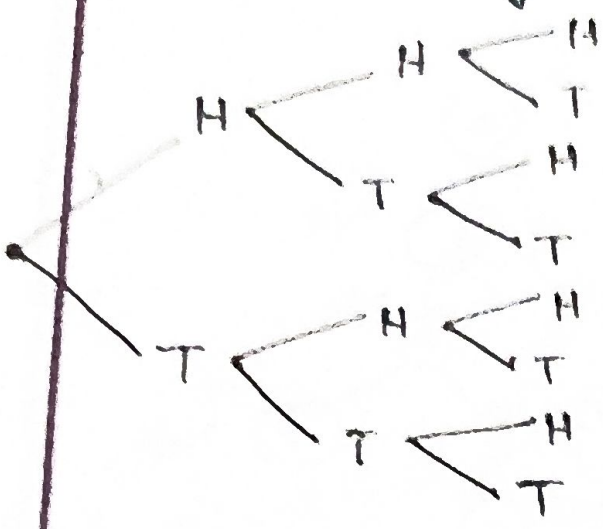
B — Toss 2: H, T

$\Rightarrow$ Independent events



$S = \{HH, HT, TH, TT\}$

Ex: Tossing a coin 3 times



Sample Space = {HHH, HHT, HTH, HTT, THH, THT, TTH, TTT}

No. of outcomes = 8

A: Getting exactly one Head,

$$P(A) = \frac{3}{8}$$

$$P(A) = \frac{{}^3C_1}{2^3} = \frac{3}{8}$$

B: Getting at least one Head

$$P(B) = \frac{{}^3C_1 + {}^3C_2 + {}^3C_3}{8} = \frac{7}{8}$$

X = number of Heads

= 0, 1, 2, 3

$$P(x=0) = \frac{{}^3C_3}{8} = \frac{1}{8}$$

$$P(x=1) = \frac{{}^3C_1}{8} = \frac{3}{8}$$

$$P(x=2) = \frac{{}^3C_2}{8} = \frac{3}{8}$$

$$P(x=3) = \frac{{}^3C_3}{8} = \frac{1}{8}$$

random variable

discrete continuo

x	$P(x=x_i)$ $i=1,\ldots,4$
x_1 0	1/8
x_2 1	3/8
x_3 2	3/8
x_4 3	1/8

Probability distribution

$P(x=x_i)$ ⇒ Probability distribution function

Random Variable ওর " " ⇒ PMF (Prob. Mass Function)

$$\boxed{P(x=x_i) = \frac{{}^3C_{x_i}}{8}}$$

Expected value = mean value

$$E[x] = \frac{\sum_{i=1}^{4} (x=x_i)\, P(x=x_i)}{\sum_{i=1}^{4} P(x=x_i)}$$

$$= \frac{0\times 1/8 + 1\times 3/8 + 2\times 3/8 + 3\times 1/8}{1/8 + 3/8 + 3/8 + 1/8}$$

$$= \frac{3}{2} = 1.5 \text{ Heads}$$

Long Run average

Q Dice roll 2 times.

x = Sum of two numbers and the sum is even.

$P(x) = ?$

PMF

* Suit: Heart, Diamond, Club, Spade

  2-10 [Num] — Heart, Diamond: red; Club, Spade: black

  9 + A, K, Q, J } Face Card

Exp: Draw a card from a standard Deck.

$n(S) = 52$

A: Drawing a king

B: " " spade

$A \cap B$ = Drawing a king of spade

$P(A) = \frac{4}{52}$ $\quad P(A \cap B) = \frac{1}{52}$

$P(B) = \frac{13}{52}$

Given that the drawn card is always a spade. What is the probability of a king being drawn.

↑ Prob $= \frac{1}{13}$

Conditional Prob,

$$P(A|B) = \frac{1}{13} = \frac{1/52}{13/52} = \frac{P(A \cap B)}{P(B)}$$

↑ Prob. of A given B has already happened

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$\Rightarrow P(B|A)\,P(A) = P(B \cap A)$$

$$\Rightarrow P(B \cap A) = P(B|A)\,P(A)$$

$$= P(A|B)\,P(B)$$

dependent event

$$P(A|B) = P(A) \quad [A \& B \text{ are independent}]$$

$$P(B|A) = P(B)$$

$$P(A \cap B) = P(A)\,P(B) = P(B)\,P(A) \quad [\text{Independent}]$$

Stat

31.8.25

* Suppose, a disease affects 1% of population. A medical test detects it with :

i) 99% chance if a person has the disease

ii) 95% " " " does not have "

iii) What is the prob. that the test result is positive ?

If A & B are independent,

$$P(A \cap B) = P(A) \times P(B)$$

If A & B are dependent

$$P(A \cap B) = P(A|B)\,P(B) = P(B|A) \cdot P(A)$$

D → A person having disease

$P(D) = 1\% = 0.01$

Complimentary Prob, $P(D' \text{ or } D^c) = 1 - P(D)$

$= 1 - 0.01 = 0.99$

A = Test result is positive

$P(A|D) = 99\% = 0.99$ → True positive result

$P(A|D^c) = 95\% = 0.95$ → False " "

$P(A) = P(D)\,P(A|D) + P(D^c) \cdot P(A|D^c)$

$= 0.01 \times 0.99 + 0.99 \times 0.95$

$= 0.9504$

ii) What is the prob that a person actually has the disease given that his test result is positive?

$$P(D|A) = P(D)\,P(A) / P(A)$$

$$= \frac{P(D \cap A)}{P(A)} = \frac{P(A|D)\,P(D)}{P(A)}$$

$$= \frac{0.99 \times 0.01}{0.9504} = 0.01042$$

P(D) → prior prob.

P(A) → posterion prob

P(A|D) → evidence based prob

→ Baye's Theorem

Q.

Bag/urn 1	Bag/urn 2
5 red balls + 7 blue balls	8 red + 9 blue balls

Exp: Draw 2 balls from either urn-1 or urn-2 What is the probability that 2 balls drawn are red?

A → ~~choosing~~ Two balls drawn (are red) from an urn

B → random selection of an urn

B_1, B_2 $\quad P(B_1) = \frac{1}{2}$

$P(B_2) = \frac{1}{2}$

$12C_2$ → Total

$5C_2$ → Favourable outcome

for B_1: $P(A|B_1) = \frac{5C_2}{12C_2}$

for B_2: $P(A|B_2) = \frac{8C_2}{17C_2}$

$P(A) = P(B_1)\,P(A|B_1) + P(B_2)\,P(A|B_2)$

$= \frac{1}{2} \times \frac{5C_2}{12C_2} + \frac{1}{2} \times \frac{8C_2}{17C_2}$

↓ Evidence

ii) What is the probability that two red balls are drawn from urn 1? / If [2 balls] (evidence) are red, what is the prob that they are drawn from urn 1?

$P(B_1) = P(B_1|A) = \frac{P(B_1 \cap A)}{P(A)}$

Posterior prob.

$= \frac{P(A|B_1)\,P(B_1)}{P(A)}$

With replacement → H.W

Without " → H.W

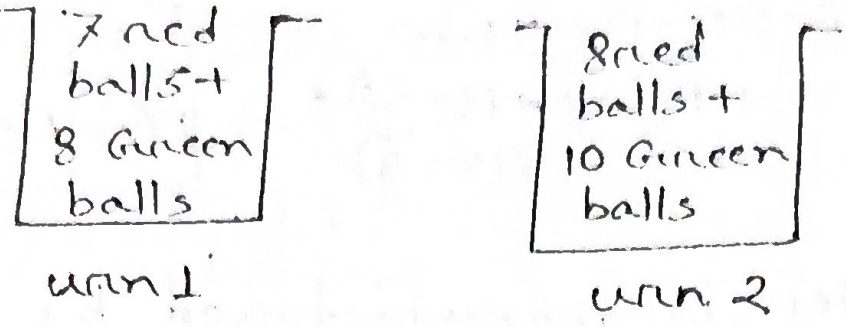
$R_1 \quad R_1$
$R_2 \quad R_2 = 3 \times 3$
$R_3 \quad R_3$

$R_1 \quad R_1 \quad R_1$
$R_2 \quad R_2 \quad R_2$
$R_3 \quad R_3 \quad R_3$

$3 \times 3 \times 3 = 3^3$

PMF (discrete)
PDF (continuous)
Prob. density fun.

Q



Exp: Draw 3 balls from either urn (without replacement).

X = The number of red balls drawn (Random variable)

i) Find the PMF of X. $X = 0, 1, 2, 3$

ii) $E[X]$

case 1 (urn 1)

$$P(X=0|_{urn1}) = \frac{{}^8C_3 \times {}^7C_0}{{}^{15}C_3}$$

$$P(X=1|_{urn1}) = \frac{{}^7C_1 \times {}^8C_2}{{}^{15}C_3}$$

$$P(X=2|_{urn1}) = \frac{{}^7C_2 \times {}^8C_1}{{}^{15}C_3}$$

$$P(X=3|_{urn1}) = \frac{{}^7C_3 \times {}^8C_0}{{}^{15}C_3}$$

case 2 (urn 2)

$$P(X=0|_{urn2}) = \frac{{}^{10}C_3 \times {}^8C_0}{{}^{18}C_3}$$

$$P(X=1|_{urn2}) = \frac{{}^8C_1 \times {}^{10}C_2}{{}^{18}C_3}$$

$$P(X=2|_{urn2}) = \frac{{}^8C_2 \times {}^{10}C_1}{{}^{18}C_3}$$

$$P(X=3|_{urn2}) = \frac{{}^8C_3 \times {}^{10}C_0}{{}^{18}C_3}$$

$$\boxed{P(A \cup B) = P(A) + P(B) - P(A \cap B)}$$

$$P(X=0) = P(\text{urn 1}) \times P(X=0|\text{urn 1}) + P(\text{urn 2}) \times P(X=0|\text{urn 2})$$
$$= \frac{1}{2} \times \frac{{}^8C_3 \times {}^7C_0}{{}^{15}C_3} + \frac{1}{2} \times \frac{{}^{10}C_3 \times {}^8C_0}{{}^{18}C_3}$$

$$P(X=1) = P(\text{urn 1}) \times P(X=1|\text{urn 1}) + P(\text{urn 2}) \times P(X=1|\text{urn 2})$$

⋮

ii) $E[x] = \frac{0 \times P(x=0) + 1 \times P(x=1) + \dots}{P(x=0) + P(x=1) + P(x=2) + P(x=3)}$

x	0	1	2	3
$P(x)$	$P(x=0)$	$P(x=1)$	$P(x=2)$	$P(x=3)$

Pmf

* Q A = Scotch is manufactured by plant 1
B = " " " " plant 2
S = A " " " a plant is standard quality

(i) $P(A|S) = \frac{P(A \cap S)}{P(S)} = \frac{P(S|A)P(A)}{P(S)}$ ← prior prob.

$P(A) = 80\%$ $\quad P(S|A) = \frac{85}{100}$

$P(B) = 20\%$ $\quad P(S'|A) = \frac{15}{100}$

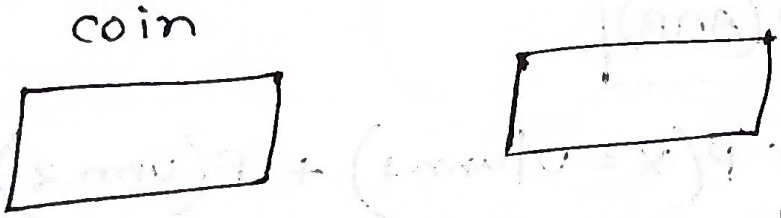
$P(S|B) = \frac{65}{100}$

$P(S'|B) = 1 - P(S/B) = \frac{35}{100}$

$P(S) = P(A) \cdot P(S|A) + P(B) \cdot P(S|B)$

* 1 special + 4 normal coin

coin



N = Normal coins randomly selected
S = coin with two heads are " "
H = Three times head in a toss

$P(S|H) = \frac{P(S \cap H)}{P(H)} = \frac{P(H|S)P(S)}{P(H)}$

Case 1 : Normal coin

$P(H|N) = \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} = \frac{1}{8}$

~~$P(H'|N) = 1 - \frac{1}{8}$~~

$P(H) = P(N)P(S|N) + P(S)P(H|S)$

T_1 = A speaks the truth

T_2 = B " " "

$P(T_2) = \frac{4}{5}$

$\therefore P(T_2') = 1 - \frac{4}{5} = \frac{1}{5}$

Case 2 : Special coin

$P(H|S) = 1 \times 1 \times 1 = 1$

$P(T_1) = \frac{2}{3}$

$P(T_1') = 1 - \frac{2}{3} = \frac{1}{3}$

B = The assertion that both A and B have agreed that the drawn ball is black

$P(B) = \frac{1}{6}$ → black ball

$P(B) = \frac{1}{6} \times \frac{2}{3} \times \frac{4}{5} + \frac{5}{6} \times \frac{1}{3} \times \frac{1}{5}$

$P(T_1 \cap T_2)|B = \frac{P((T_1 \cap T_2) \cap B)}{P(B)} = \frac{1/6 \times 2/3 \times 4/5}{1/6}$

Success → Functional

Failure → non-functional / defective

RE-1: { H, T }

X_1 = Success = H = 1

Failure = T = 0

Bernoulli's Prob. Distribution

$x = x_1$	$P(x = x_i)$
1	1/2
0	1/2

Success = P

Failure = 1 − P

$X = 1, 0$ $\quad X \sim \text{Bernoulli}(P)$

$P(X=1) = P$

$P(X=0) = 1-P$ } PMF

$P(X=k) = p^k (1-p)^{1-k}$

$P(X=1) = p^1 (1-p)^0$ } Compact form pmf of bernoulli's distribution

$P(X=0) = 1-p$

Y = 5 on the face of the dice

$P(Y=5) = \frac{1}{5}$

$E[X] = 1 \cdot P + 0 \times (1-P)$

$\Rightarrow E[X] = P$

$\bar{x}$ or $\mu = E[X] = P$

~~Var(x) =~~

$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n}$

for population,

$\sigma^2 = \frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}$

$$\text{Var}(X) = \text{expected value/average of } (x-\mu)^2$$

$$= E[(x-\mu)^2] = E[x^2 - 2x\mu + \mu^2]$$

$$= E[x^2] - E[2x\mu] + E[\mu^2]$$

$$= E[x^2] - 2\mu E[x] + \mu^2$$

$$= E[x^2] - 2[E[x]]^2 + (E[x])^2$$

$$\boxed{\text{Var}(X) = E[x^2] - (E[x])^2}$$

For any bernoulli random variable,

$$E[x] = p$$

$$E[x^2] = 1^2 \cdot p + 0^2(1-p)$$

$$= p$$

$x = k$	$x^2 = k^2$	$P(x=k)$
1	1^2	p
0	0^2	$1-p$

$$\text{Var}(X) = p - p^2 = p(1-p)$$

Bernoulli Trial => 1 execution of Bernoulli random experiment.

Q X = no of heads of you toss a coin

(i) Create pmf

(ii) Compute $E[X]$ and $\text{Var}[X]$

$X = H, T$

$1, 0$

$X = K$	$P(X=K)$
1	1/2
0	1/2

(ii) $E[X] = 1 \cdot 1/2 + 0 \times 1/2$

$= 0.5$

$Var(X) = E[X^2] - (E[X])^2$

$E[X^2] = 1^2 \times \frac{1}{2} + 0^2 \times \frac{1}{2} = 0.5$

$Var(X) = 0.5 - (0.5)^2 = 0.25$

Q2

(A)	(B) $10. win
(C)	(D) $100 loose

1. Create PMF.
2. Compute $E[X]$ and $Var[X]$

1. X = amount of money profit

$X = k$	$P(X = k)$
10	1/4
-100	3/4

2. $E[X] = 10 \times \frac{1}{4} + (-100) \times \frac{3}{4} = -0.7 - 72.5$ $

$E[X^2] = 10^2 \times \frac{1}{4} + (-100)^2 \times \frac{3}{4} =$

$Var(X) = E[X^2] - (E[X])^2 =$

$n = 4$

H T x_1 | H T x_2 | H T x_3 | H T x_4

$P = \frac{1}{2}$ $P = \frac{1}{2}$ $P = \frac{1}{2}$ $P = \frac{1}{2}$

q q q q

n Bernoulli trials
Every " " is independent
The success prob P or failure prob remains constant

Binomial Experiment = Combination of n Bernoulli trial trials

$n = 4$,
Tossing a coin 4 times

$X = X_1 + X_2 + X_3 + X_4 + \cdots + X_n$

⇓

Binomial random variable

No. of trials, $n = 4$;

How many times you can get $k = 2$ success/Head

$X = \underbrace{0, 1, 2, 3, 4}_{k}$

$P(X = 2)$

$P(X = k)$

Case 1:

H H T T
P P p^2q^2 q q

Case 2:

H T H T
P q P q
p^2q^2

Case 3

H T T H
P q q P
p^2q^2

Case 4

T H H T
q P P q
p^2q^2

Case 5:

T H T H
q P q P
p^2q^2

Case 6

T T H H
q q P P
p^2q^2

$G\ p^1 q^2$

$$\binom{4}{2} p^1 q^2 = \binom{4}{2} p^k q^{n-k} = \binom{n}{k} p^k q^{n-k}$$

$$\boxed{P(x=k) = \binom{n}{k} p^k q^{n-k}} \Rightarrow \text{pmf of binomial dist}$$

$n = 8$

$$P(x=3) = \binom{8}{3} p^3 (1-p)^{8-3}$$

ii) What is the prob that you can get atmost 4 success/heads?

$$P(X \leq 4) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)$$

$\Downarrow$

CDF (Cumulative Dist. Func.)

$$P(X \leq K) = P(x=0) + P(x=1) + \cdots + P(x=K-1) + P(x=K)$$

$$= \binom{8}{0} \left(\frac{1}{2}\right)^0 \left(\frac{1}{2}\right)^{8-0} + \binom{8}{1} \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^{8-1} + \cdots$$

iii) $P(X > 4) = 1 - P(X \leq 4)$

$X \sim \text{Binomial}(n, k$

$\downarrow$

Binomial R.V

$$E[x] = (x=0)\,P(x=0) + (x=1)\,P(x=1) + \cdots + (x=k)\,P(x=k)$$

$$= \sum_{k=0}^{k} (x=k)\,P(x=k)$$

$$X = X_1 + X_2 + \cdots + X_n$$

$$E[x] = E[X_1 + X_2 + \cdots + X_n]$$

$= E[X_1] + E[X_2] + \cdots + E[X_n]$

$= p + p + \cdots + p$

$= np$

$\text{Var}(X) = E[X^2] - (E[X])^2$

$\text{Var}(X) = \text{Var}(X_1 + X_2 + \cdots X_n)$

$= \text{Var}(X_1) + \text{Var}(X_2) + \cdots + \text{Var}(X_n)$

$= p(1-p) + p(1-p) + \cdots p(1-p)$

$= np(1-p)$

Q ~~x~~ $k = 3$. Biased with success prob $= 0.7$. A coin is tossed many times, and you have got $k = 3$ successes (heads). How many trials required to get k success.

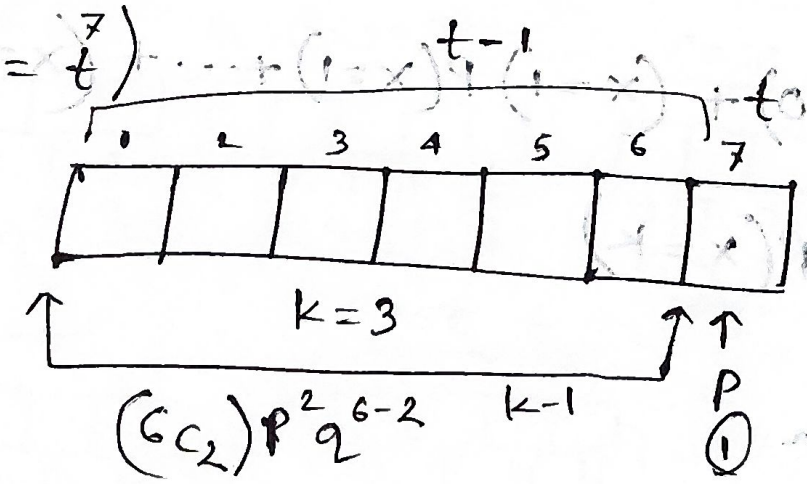
Success prob. of one trial, $p = ~~1/2~~ 0.7$

Failure " " " ", $q = 1 - p = 0.3$

$X = k, k+1, k+2, \cdots, t$

$P(X = t)$

$P(X = 7)$



$(^6C_2) p^2 q^{6-2}$ (1)

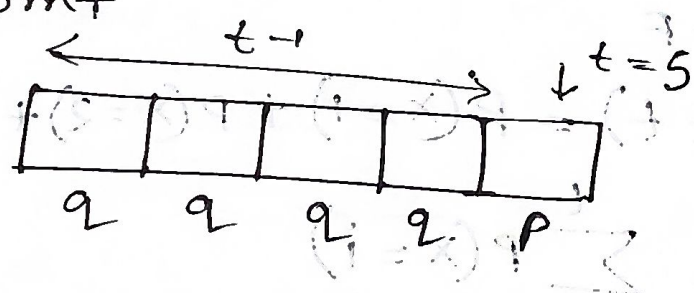
$$P(X=t) = \binom{t-1}{k-1} p^{k-1} q^{(t-1)-(k-1)} \cdot p$$

$$\boxed{P(X=t) = \binom{t-1}{k-1} p^{k} q^{t-k}} \Rightarrow \text{PMF. Negative Binomial Dist}$$

~~P(x=~~ * What is the prob to obtain 3 successes in first 7^{th} trials ?

compute $E[X]$, $Var[X]$ ⇒ Home task

Q A biased coin is tossed with S.P = 0.6. How many trials required to get the first success? Design pmf



* Geometric Distribution

What is the prob to get first success in 20^{th} attempt ?

$$P(X=t) = q^{t-1} \cdot p$$

$$P(X=20) = (0.4)^{20-1} \cdot 0.6$$

$p = 0.6$
$q = 0.4$

Compute $E[X]$, $Var[X]$; if x is a geometric. r.v

Geometric R.D: First success in t^{th} trial

$$E[X] = (X=1)P(X=1) + (X=2)P(X=2) + \cdots + (X=t)P(X=t)$$

X = Geometric R.V ; Num of trials to get first success

$$E[X] = \sum_{i=1}^{t} (X=i)\,P(X=i)$$

$$= \sum_{i=1}^{t} i\,P(X=i) = \sum_{i=1}^{t} i \cdot q^{i-1} \cdot p$$

$$= p \sum_{i=1}^{t} i q^{i-1}$$

$$\therefore E[X] = p\left[1 \times q^{1-1} + 2q^{2-1} + 3q^{3-1} + \cdots + tq^{t-1}\right]$$

$$= p\left[1 + 2q + 3q^2 + \cdots + tq^{t-1}\right] \Rightarrow \text{H.W}$$

$\text{Var}(X) = ?$ CDF

$$P(X \le t) = P(X=1) + P(X=2) + \cdots + P(X=t)$$

$$= \sum_{i=1}^{t} P(X=i)$$

$$= \sum_{i=1}^{t} q^{i-1} p = p \sum_{i=1}^{t} q^{i-1}$$

$$= p\left[q^0 + q^1 + q^2 + \cdots + q^{t-1}\right]$$

$$= p\left(\frac{1-r^t}{1-r}\right)$$

$$\therefore P(X \le t) = p\left(\frac{1-q^t}{1-q}\right) = p\left(\frac{1-q^t}{1-(1-p)}\right)$$

$$= 1 - q^t = 1 - (1-p)^t$$

Stat.

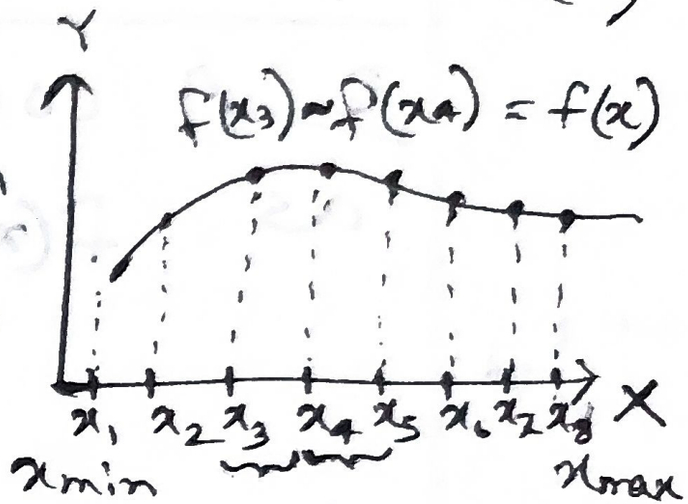
15-10-25

Continuous R.V

Q Guess some continuous R.Vs

i) Height
ii) Weight
iii) BP
iv) Age
v) IQ score
vi) Cgpa

$x \to f(x)$



$x_{min} = 4.10$ feet ~ 147 cm

$x_{max} = 6.20$ feet ~ 183 cm

$x_{min} \leq (x = x) \leq x_{max}$

$x_3 \leq (x = x) \leq x_4$

$\Delta x = x_4 - x_3$

$A = f(x)\,dx$

x = denotes the height of every person of BD

$X = x/K$

$X = x$ $f(x) \rightarrow$ for the time being consider it as freq

$P(X = x_4) = \frac{1}{\infty} = 0$

Total outcome $= \infty$

$P(X = x) = 0$

$f(x) \rightarrow$ probability density fun (PDF)

$f(x)\,dx \rightarrow$ prob

$P(x_5 \leq (X = x) \leq x_6) = f(x)\,dx$

$P(x_2 \leq (X = x) \leq x_7) = \int_{x_2}^{x_7} f(x)\,dx$

$P(a \leq (X = x) \leq b) = \int_a^b f(x)\,dx$

Properties of Prob Dist of continuous R.V

i) $f(x) \geq 0$ where $f(x)$ represents $f(x)$

ii) $P(x_{min} \leq X = x \leq x_{max}) = \int_{x_{min}}^{x_{max}} f(x)\,dx = 1$

iii) $P(a \leq x \leq b) = \int_a^b f(x)\,dx$

Uniform Distribution!

Q A continuous R.V X is defined with the PDF as

$$f(x) = \begin{cases} \frac{1}{x_{max} - x_{min}} ; x_{min} \leq (X = x) \leq x_{max} \\ 0 \quad \text{otherwise} \end{cases}$$

i) Compute $P(a \leq X \leq b)$

ii) " $E[X]$

iii) " $Var[X]$

i) $P(a \le x \le b) = \int_a^b f(x)\,dx$

ii) $E[X] = \int_a^b x f(x)\,dx$

$$= \int_a^b x \frac{1}{x_{max} - x_{min}}\,dx$$

$$= \frac{1}{b-a} \int_a^b x\,dx$$

$$= \frac{1}{b-a} \left[\frac{x^2}{2}\right]_a^b$$

$$= \frac{1}{2(b-a)} (b^2 - a^2) = \frac{(b+a)(b-a)}{2(b-a)}$$

$$= \frac{a+b}{2} = \frac{x_{max} + x_{min}}{2}$$

iii) $Var[X] = E[X^2] - (E[X])^2$

$$E[X^2] = \int_{x_{min}}^{x_{max}} x^2\,dx\, f(x)\,dx$$

$$= \frac{1}{x_{max} - x_{min}} \int_{x_{min}}^{x_{max}} x^2\,dx$$

$$= \frac{1}{x_{max} - x_{min}} \left[\frac{x^3}{3}\right]_{x_{min}}^{x_{max}}$$

$$= \frac{1}{3(x_{max} - x_{min})} (x_{max}^3 - x_{min}^3)$$

$$= \frac{(x_{max} - x_{min})(x_{max}^2 + x_{max} \cdot x_{min} + x_{min}^2)}{3(x_{max} - x_{min})}$$

$$= \frac{x_{max}^2 + x_{max}x_{min} + x_{min}^2}{3}$$

$$Var[X] = \frac{x_{max}^2 + x_{max}x_{min} + x_{min}^2}{3} - \left(\frac{x_{max}+x_{min}}{2}\right)^2$$

$$= \frac{x_{max}^2 + x_{max}x_{min} + x_{min}^2}{3} - \frac{x_{max}^2 + 2x_{max}x_{min} + x_{min}^2}{4}$$

$$= \frac{(x_{max} - x_{min})^2}{12}$$

* $a = 0,\ b = 4$

$$f(x) = \frac{1}{b-a} = \frac{1}{4}$$

$$f(x) = \begin{cases} \frac{1}{4}, & 0 \le x \le 4 \\ 0 & \text{elsewhere} \end{cases}$$

ii) $$P(x \ge 3) = \int_3^4 f(x)\,dx$$

$$= \int_3^4 \frac{1}{4}\,dx = \frac{1}{4}[x]_3^4$$

$$= \frac{1}{4}(4-3) = \frac{1}{4}$$

Stat

16-10-25

Q $T_{alg} = 30$ | $T_{stat} = 20$

$A_{alg} = 25$ | $A_{stat} = 17$

continuous → inclusive

Gaussian

PDF of X, $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

μ = mean height of population

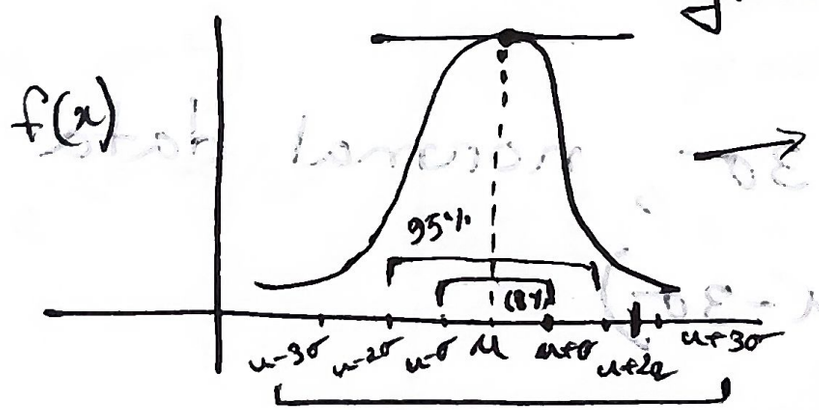
x = Normal random variable

x = any particular height

σ^2 = variance of the given height DS

σ = S.D

gradient = 0



→ Bell-shaped curve

Normal range (99.7% data)

$n\sigma = x - \mu$

$\sigma = \frac{x-\mu}{n}$

x will be mode for which,

$$\frac{d}{dx}[f(x)] = 0$$

$$\Rightarrow \frac{1}{\sqrt{2\pi\sigma^2}} \frac{d}{dx}\left[e^{-\frac{(x-\mu)^2}{2\sigma^2}}\right] = 0$$

$$\Rightarrow e^{-(x-\mu)^2/2\sigma^2} \frac{d}{dx}\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] = 0$$

$$\Rightarrow \frac{1}{2\sigma^2} 2(x-\mu)\frac{d}{dx}(x-\mu) = 0$$

$$\Rightarrow x-\mu = 0 \quad \therefore x = \mu$$

Mode is, $x = \mu$

Mean = Median = Mode = μ [Normal Dist]

Symmetric

$\sigma = x - \mu$

$2\sigma = x - \mu$

$3\sigma = x - \mu$

$n\sigma = x - \mu$

$\therefore \sigma = \frac{x-\mu}{n}$

$$Z = \frac{x-\mu}{\sigma}$$

$x = 188$

$\mu = 160$

188

$Z = 2.8$ means # is 2 SD above the mean

$\mu = 20$

$\mu - \sigma$ $\quad$ $\mu + \sigma$

15 $\quad$ 25

$\mu - 3\sigma \le x \le \mu + 3\sigma$, normal data

if $(x > \mu + 3\sigma \;||\; x < \mu - 3\sigma)$:

x = outlayer

68-95-99.7 Rule

Q

CSE	phy
$\mu_{CSE} = 3.0$	$\mu_{phy} = 3.30$
$\sigma_{CSE} = 0.40$	$\sigma_{phy} = 0.4$
$A_{CSE} = 3.70$ 2.8	$B_{phy} = 3.85$ 3.2

$$Z_A = \frac{x-\mu}{\sigma} = \frac{3.70-3.0}{0.40} = 1.75$$

$$Z_B = \frac{3.85-3.30}{0.4} = 1.375$$

$$Z_A = \frac{2.8-3.0}{0.4} = -0.5$$

$$Z_B = \frac{3.2-3.30}{0.4} = -0.25$$

Standardization Technique

* To make any D.S standard, compute z-score of that DS
for any normal x,

$$\boxed{-3 \le z \le 3}$$

If for any x, $z > 3$ or $z < -3$

x = outlayer

Normal Dist

$$X \sim N(x, \mu, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

Standard Normal Dist

$$X \sim N(x, \mu=0, \sigma^2=1)$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

Z → z-score ds

z → any particular value

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

pdf S.N.D

—x—